深圳瑞波光电子有限公司

RB-APD500-905-SMD-B2 规格书

Shenzhen Raybow Optoelectronics Co., Ltd丨深圳瑞波光电子有限公司
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修订记录						
版本	修订日期	修订内容	修订	审核	审批	生效日期
V1.0	N/A	初版	祝隆平	张春杨	胡海	2024/3/25

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Avalanche Photo Diode in SMD package 雪崩光电二极管器件 RB-APD500-905-SMD-B2

■ Applications ｜ 应用

- **LiDAR/ Laser ranging** 激光雷达/测距
- **High speed photometry** 高速光度计
- **High speed optical communications** 高速光通信
- **Medical equipment** 医疗设备

■ Features ｜ 产品特点

- **Active area diameter 0.5mm** 光敏面直径 **0.5mm**
- **QE>80%@905nm** 量子效率**>80%@905nm**
- **Low voltage, high gain, low capacitance, low noise** 低电压、高增益、低结电容、低噪声
- 可内置 **905nm** 窄带滤光片 **905nm narrow-band filter available**

■ Absolute Maximum Ratings(T_{rt}=25℃) 极限值

Parameter 参数	Symbol 符号	Min 最小值	Max 最大值	Unit 单位
Peak current 峰值电流	I_p		0.25	mA
Operating temp 工作温度	T_{op}	-40	100	℃
Storage temp 贮存温度	T_{stg}	-55	150	℃
Gain 增益	M_{max} (I_p =1nA)	200		

■ Characteristics (T_{rt}=25 ℃) 特性参数值

Parameter 参数	Symbol 符号	Test Conditions 测试条件	Min 最小值	Typ 典型值	Max 最大值	Unit 单位
Active area 有效面积	D		直径 500			μm
	S		0.196			mm^2
Responsivity 响应度	Re	M=100, λ=905 nm	53	56	60	A/W
Dark current 暗电流	I_D	M=100		0.5	1	nA
Capacitance 电容	C	M=100		1.2		pF
Rise time 上升时间	T_R	M = 100, λ = 905 nm, R_L = 50 Ω		0.4		ns
Breakdown Voltage 击穿电压	V_{BR}	I_R=2μA	130		150	V
Temperature Coefficient 温度系数			1.25	1.4	1.55	V/°C

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■ Characteristics (T_{rt}=25 ℃) 特性曲线

Figure 1: Responsivity
光谱响应度曲线(M=100)

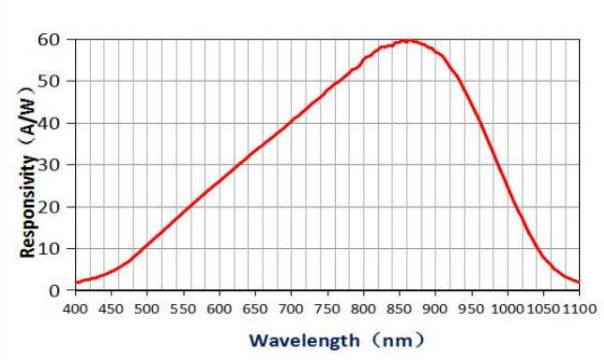


Figure 2: 905nm Filter band pass
905nm 滤光片光谱

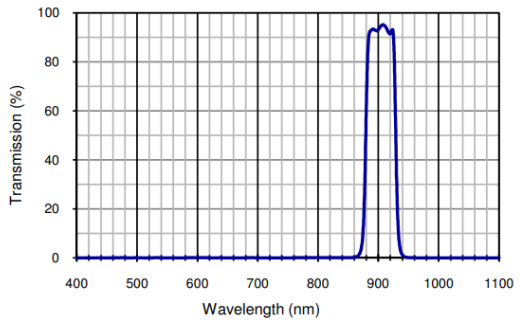


Figure 3: Quantum efficiency 量子效率 (25℃)

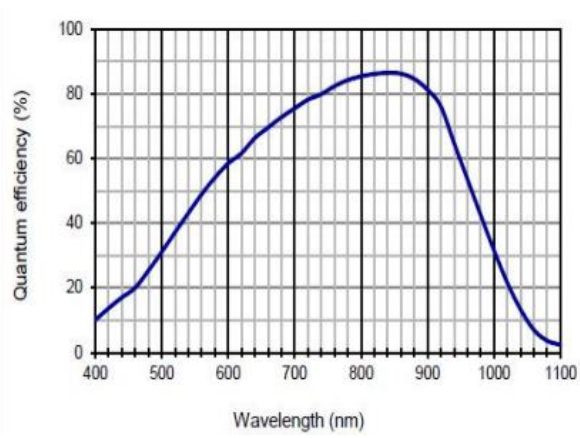


Figure 4: Capacitance 电容(25℃)

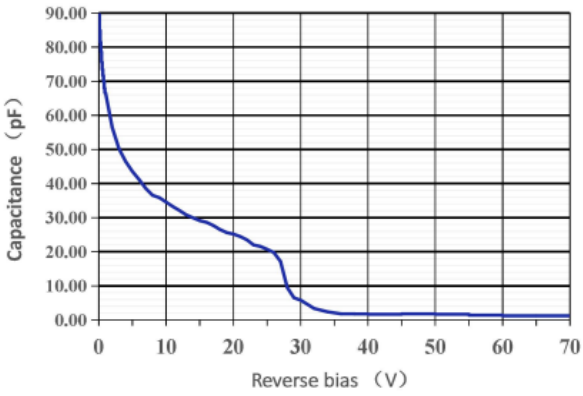


Figure 5: Gain-reverse voltage
增益-反向电压曲线

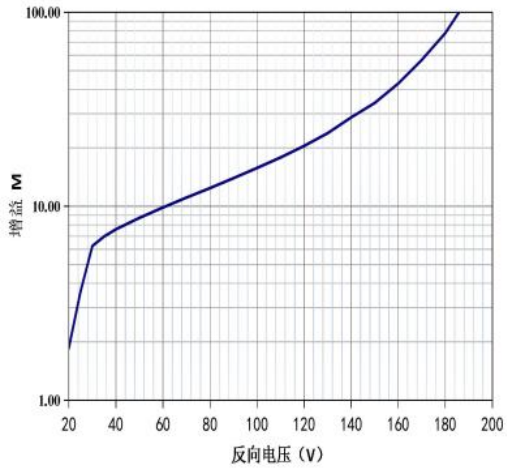
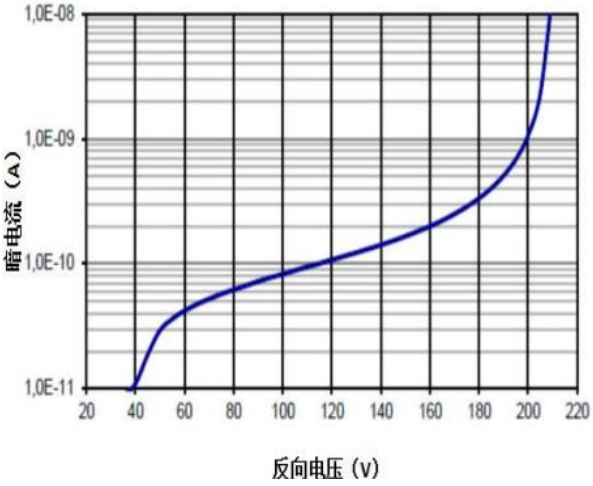


Figure 6: Dark current-reverse voltage
暗电流-反向电压曲线



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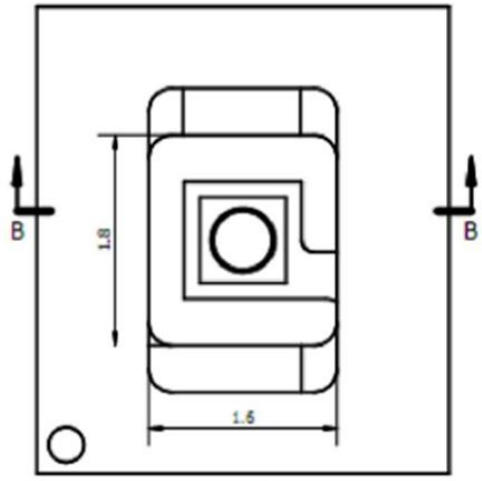
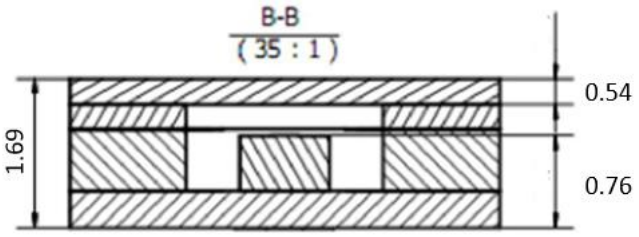
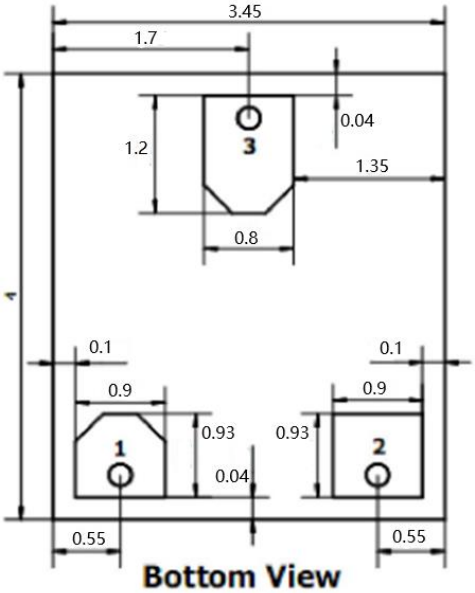
■ Dimensional Drawing 外观、尺寸及引脚定义



Top View





Bottom View

1:Anode
2:Cathode
3:NC
Unit:mm

■ Notes 注意事项

a. Specifications are subject to change without notice. No liability is assumed as a result of their use or application without confirming with our company.
此份规格书会在未通知的情况下进行更改。未与我司确认，参考此份规格书造成的使用问题，我司不承担责任。

b. Electrostatic discharge causes great damage to the device. Make sure the human body, test instruments, inspection facilities and working surfaces are well grounded during the handling and operation of the device.
静电对器件有极大伤害，在搬运以及使用中要保证人体、测试仪表、检验装置及工作台接地良好。

c. The power should be connected to a voltage regulating unit, and make sure no impulsive voltage is generated during the power on/off stage. Otherwise, the device may be damaged.
电源需要配有稳压装置，防止在开关电源过程中产生冲击电压损害器件。

d. For low light level applications, ambient light should be blocked.
对于探测光较弱的应用场景，应当遮挡环境光的照射。

e. For high gain applications, bias voltage should be temperature compensated.
对于高增益的应用场景，反偏电压需要做温度补偿。

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